Tutorial – 14

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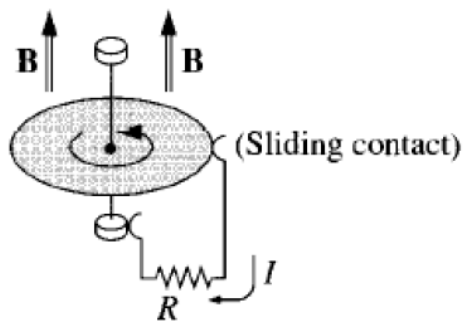
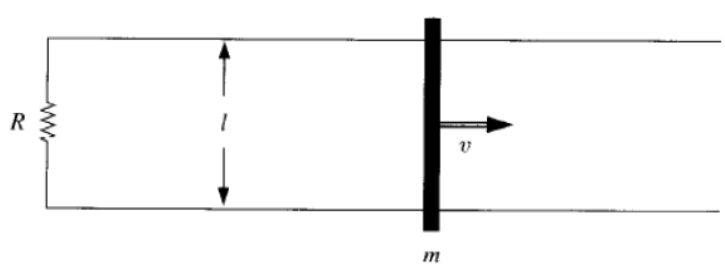
1. A metal disk of radius 'a' rotates with angular velocity ω about a vertical axis, through a uniform field **B** pointing up. A circuit is made by connecting one end of the resistor to the axle and the other end to a sliding contact, which touches the outer edge of the disk. Find the current in the resistor.



2. A metal bar of mass 'm' slides frictionlessly on two parallel conducting rails a distance l apart. A resistor is connected across the rails, and a uniform magnetic field **B**, pointing into the page, fills the entire region.



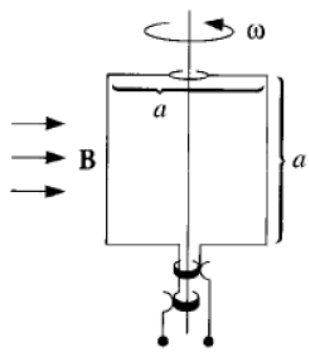
(a) If the bar moves to the right at speed v, what is the current in the resistor? In what direction does it flow?

(b) What is the magnetic force on the bar due to current flow? In what direction?

(c) If the bar starts out with speed v_0 at time $t = 0$, and is left to slide, what is its speed at a later time t?

(d) The initial kinetic energy of the bar was, of course, $\frac{1}{2}mv_0^2$. Check that the energy delivered to the resistor is exactly $\frac{1}{2}mv_0^2$.

3. A square loop (side a) is mounted on a vertical shaft and rotated at angular velocity ω. A uniform magnetic field **B** points to the right. Find the induced emf $\varepsilon(t)$ for this alternating current generator

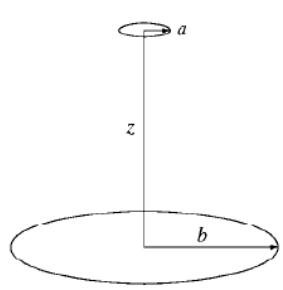


4. A long solenoid with radius 'a' and n turns per unit length carries time-dependent current $I(t)$ in the $\hat{\varphi}$ direction. Find the electric field (magnitude and direction) at a distance 's' from the axis (both inside and outside the solenoid), in the quasistatic approximation.

5. An alternating current $I = I_0 \cos(\omega t)$ flows down a long straight wire, and returns along a co-axial conducting tube of radius 'a'. Assuming that the fied goes to zero as $s \to \infty$, find **E**(s,t)

6. A small loop of wire (radius a) is held at a distance z above the center of a large loop (radius b) as shown. The planes of the two loops are parallel, and perpendicular to the common axis.

(a) Suppose the current I flows in the big loop. Find the flux through the little loop. (The little loop is so small that you may consider the field of the big loop to be essentially constant)

(b) Suppose current I flows in the little loop. Find the flux through the big loop. (The little loop is so small that you may treat it as a magnetic dipole)

(c) Find the mutual inductances, and confirm that $M_{12} = M_{21}$